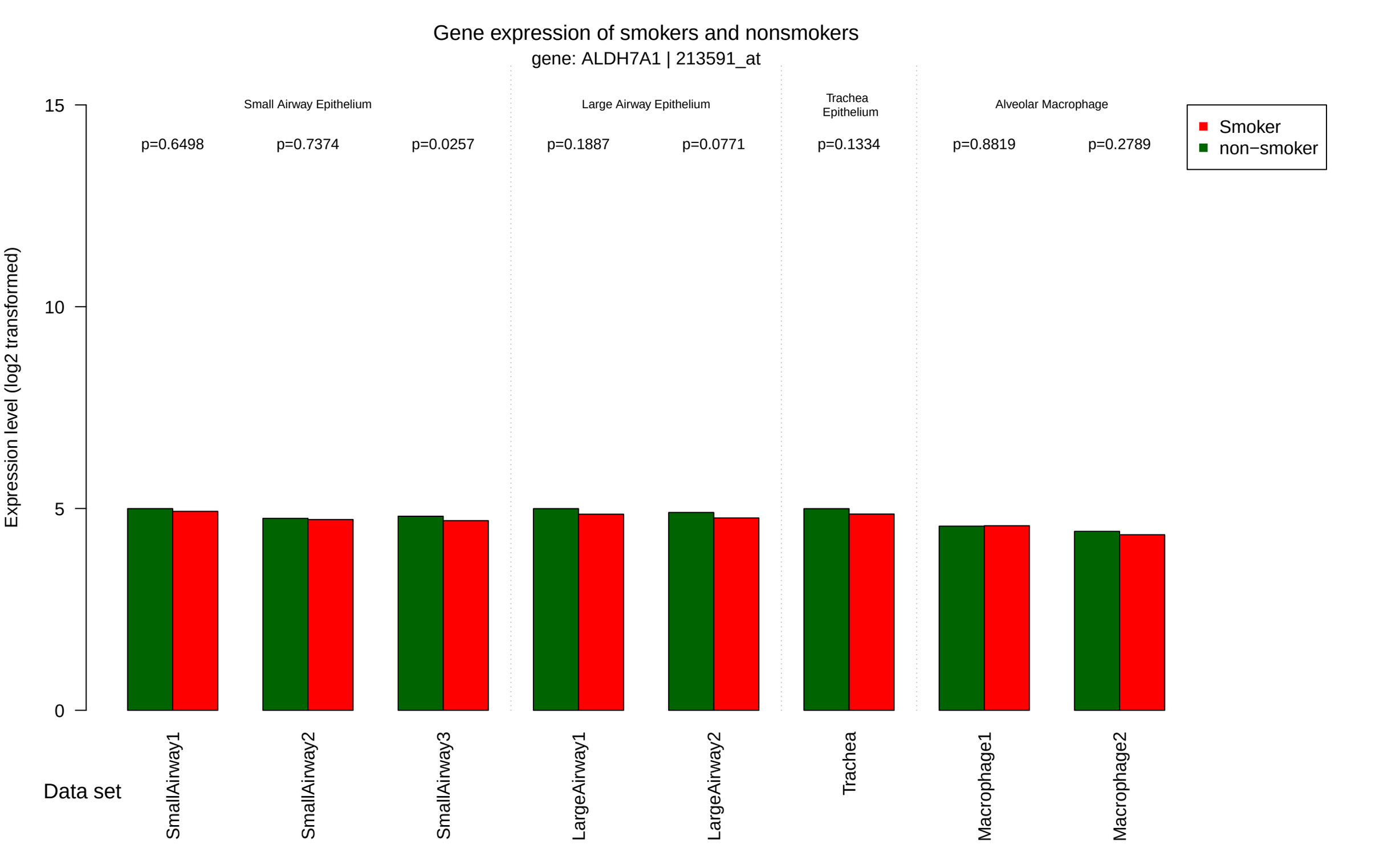
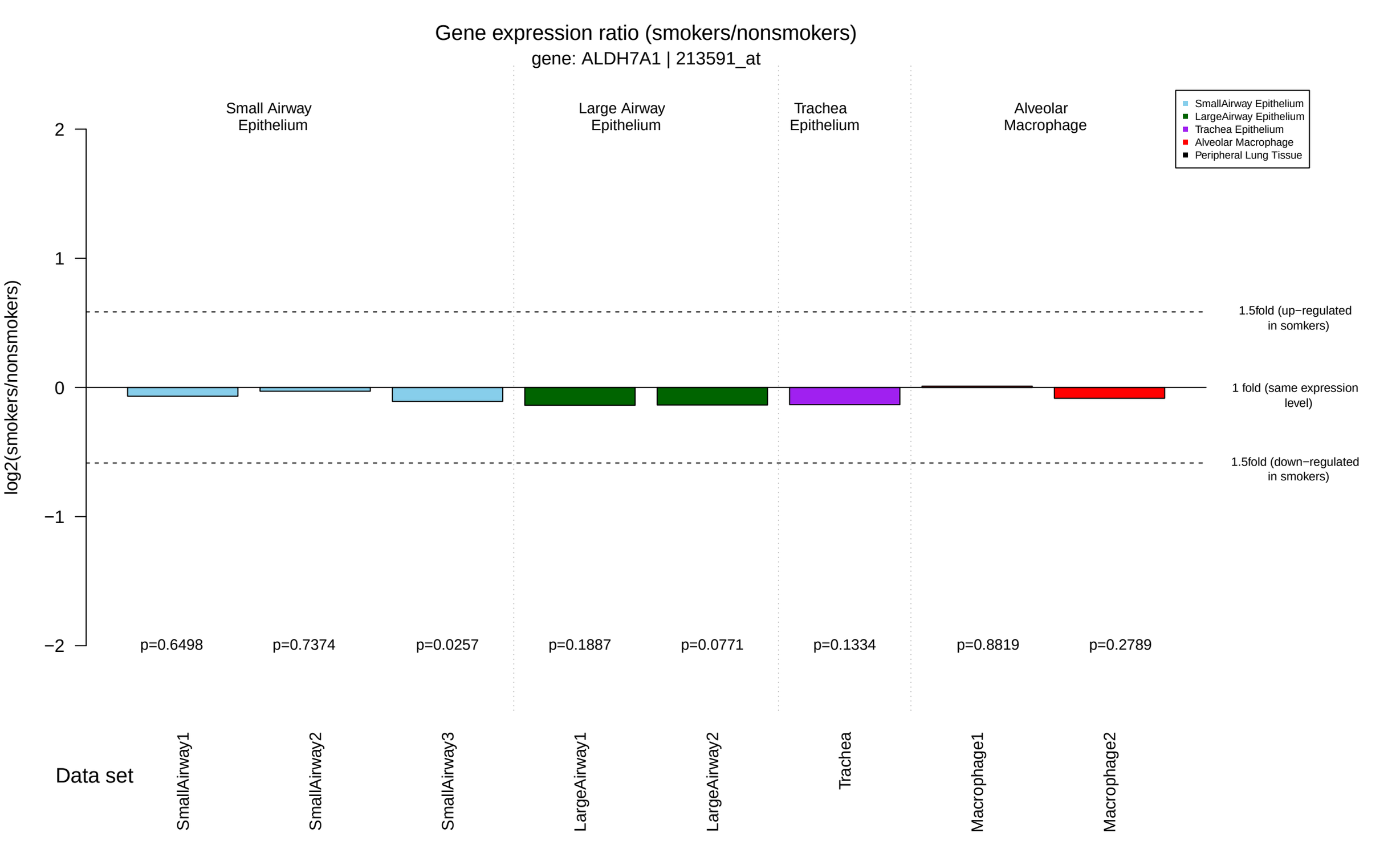
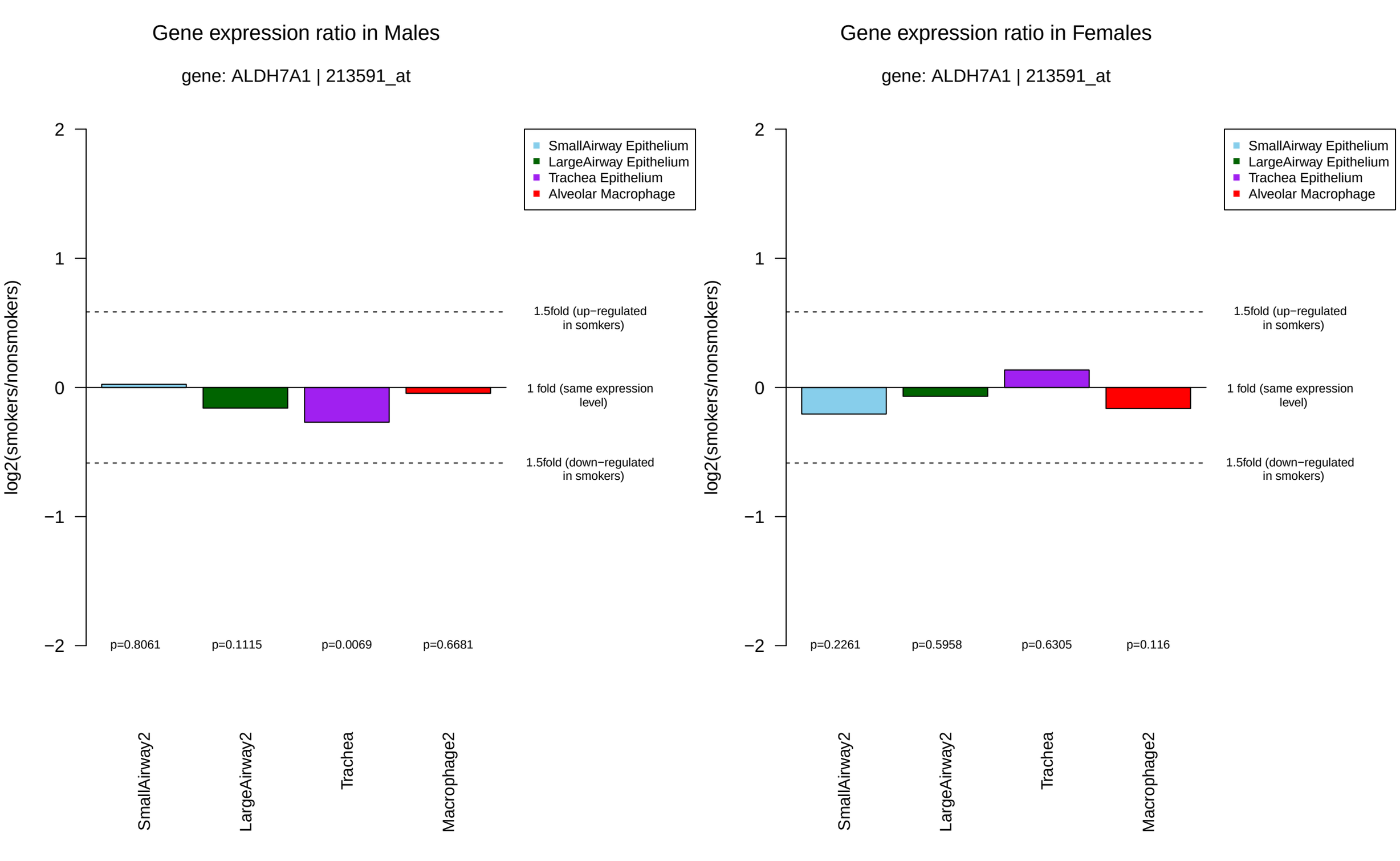
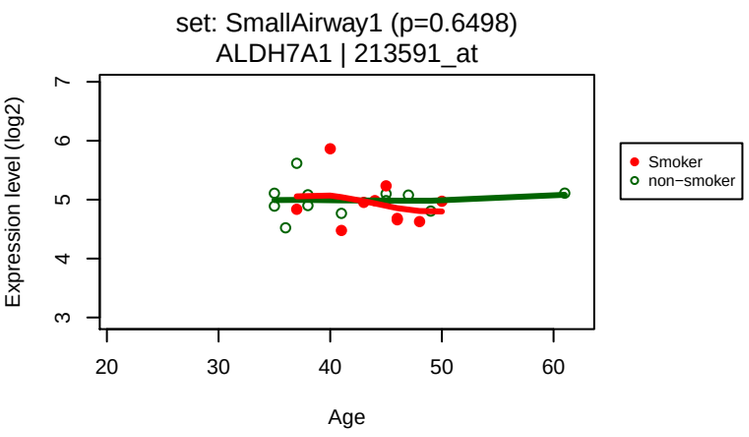
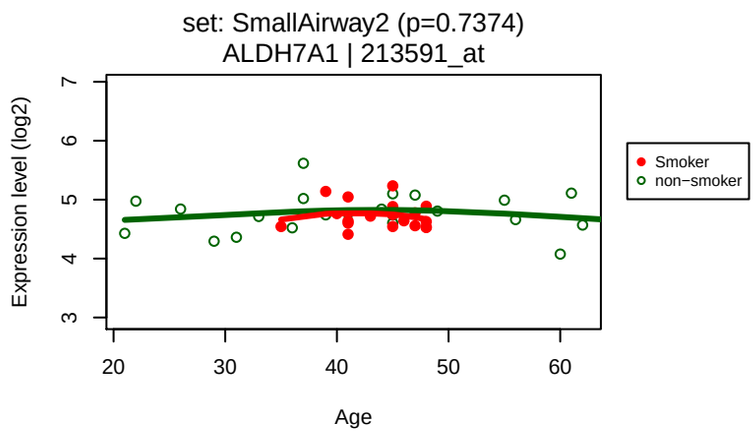
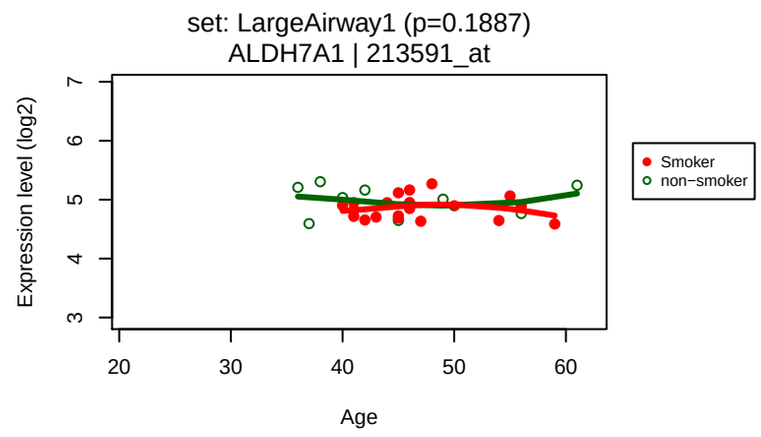

Gene expression of smokers and nonsmokers
gene: ALDH7A1 | 213591_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6498
p=0.7374
p=0.0257
p=0.1887
p=0.0771
p=0.1334
p=0.8819
p=0.2789
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

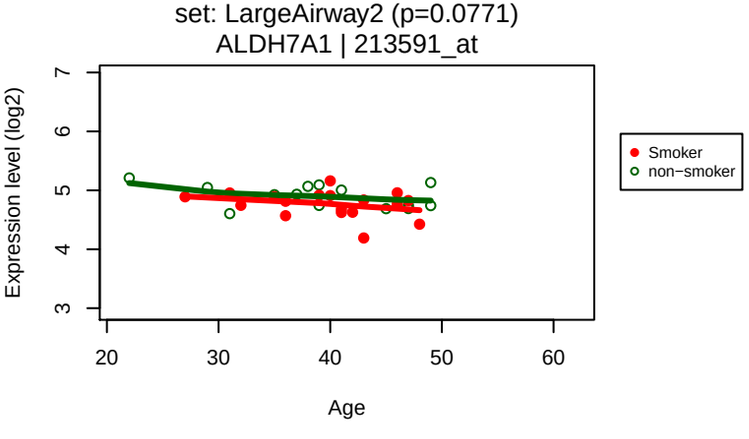

Gene expression ratio (smokers/nonsmokers)
gene: ALDH7A1 | 213591_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6498
p=0.7374
p=0.0257
p=0.1887
p=0.0771
p=0.1334
p=0.8819
p=0.2789
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

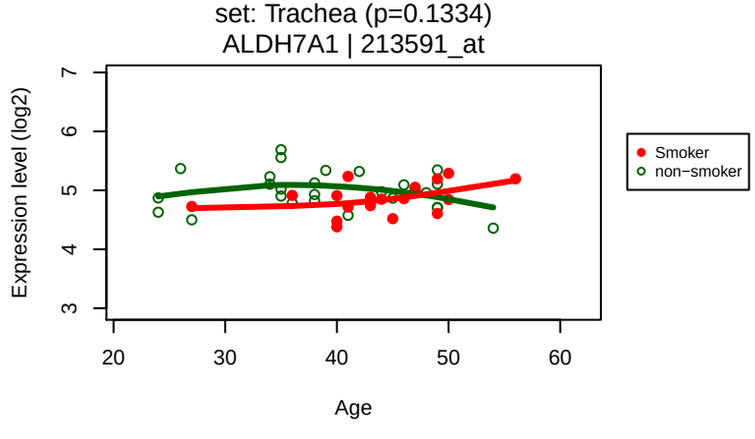

Gene expression ratio in Males
gene: ALDH7A1 | 213591_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8061
p=0.1115
p=0.0069
p=0.6681
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALDH7A1 | 213591_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2261
p=0.5958
p=0.6305
p=0.116
SmallAirway2
LargeAirway2
Trachea
Macrophage2

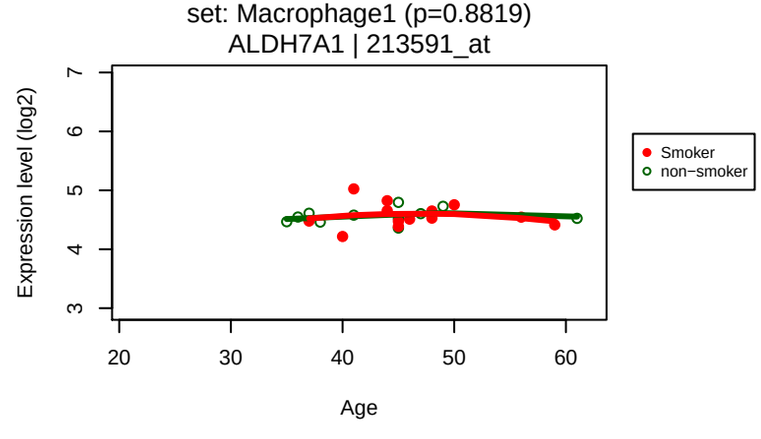

set: SmallAirway1 (p=0.6498)
ALDH7A1 | 213591_at
Expression level (log2)
Age
Smoker
non−smoker

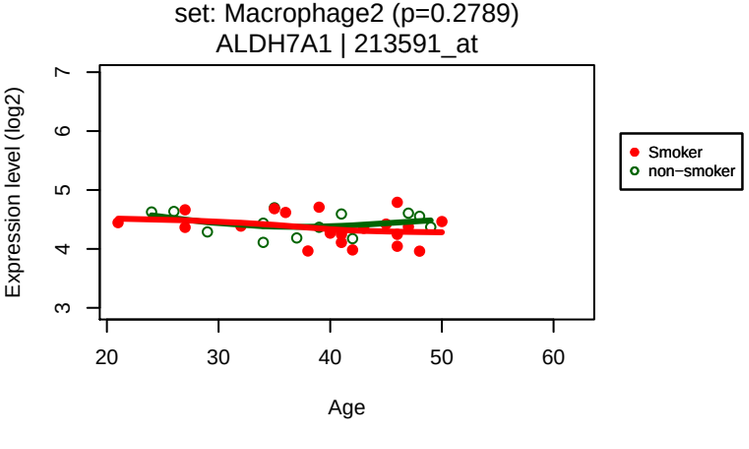

set: SmallAirway2 (p=0.7374)
ALDH7A1 | 213591_at
Expression level (log2)
Age
Smoker
non−smoker

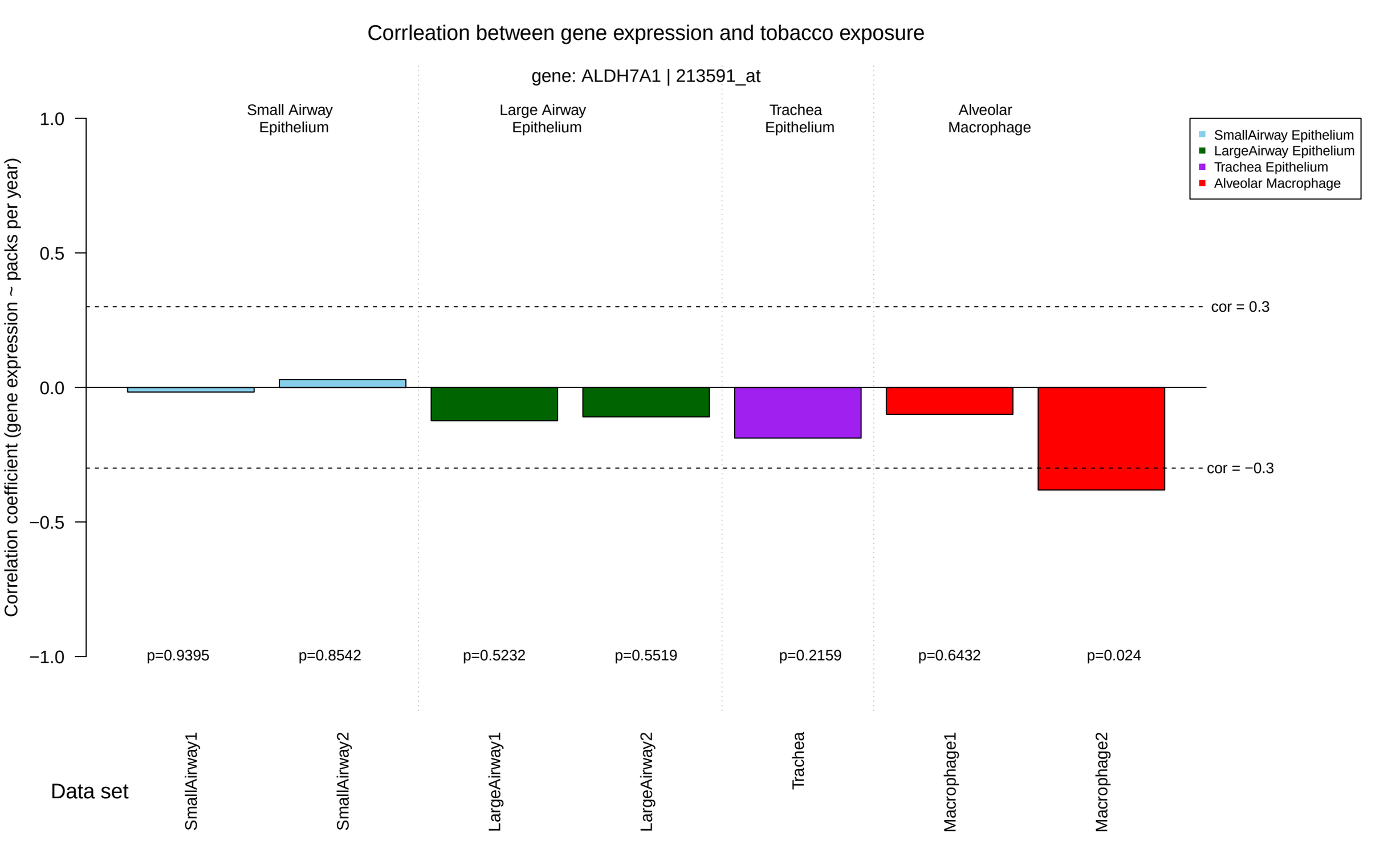

set: LargeAirway1 (p=0.1887)
ALDH7A1 | 213591_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0771)
ALDH7A1 | 213591_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1334)
ALDH7A1 | 213591_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8819)
ALDH7A1 | 213591_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2789)
ALDH7A1 | 213591_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALDH7A1 | 213591_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.9395
p=0.8542
p=0.5232
p=0.5519
p=0.2159
p=0.6432
p=0.024
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2